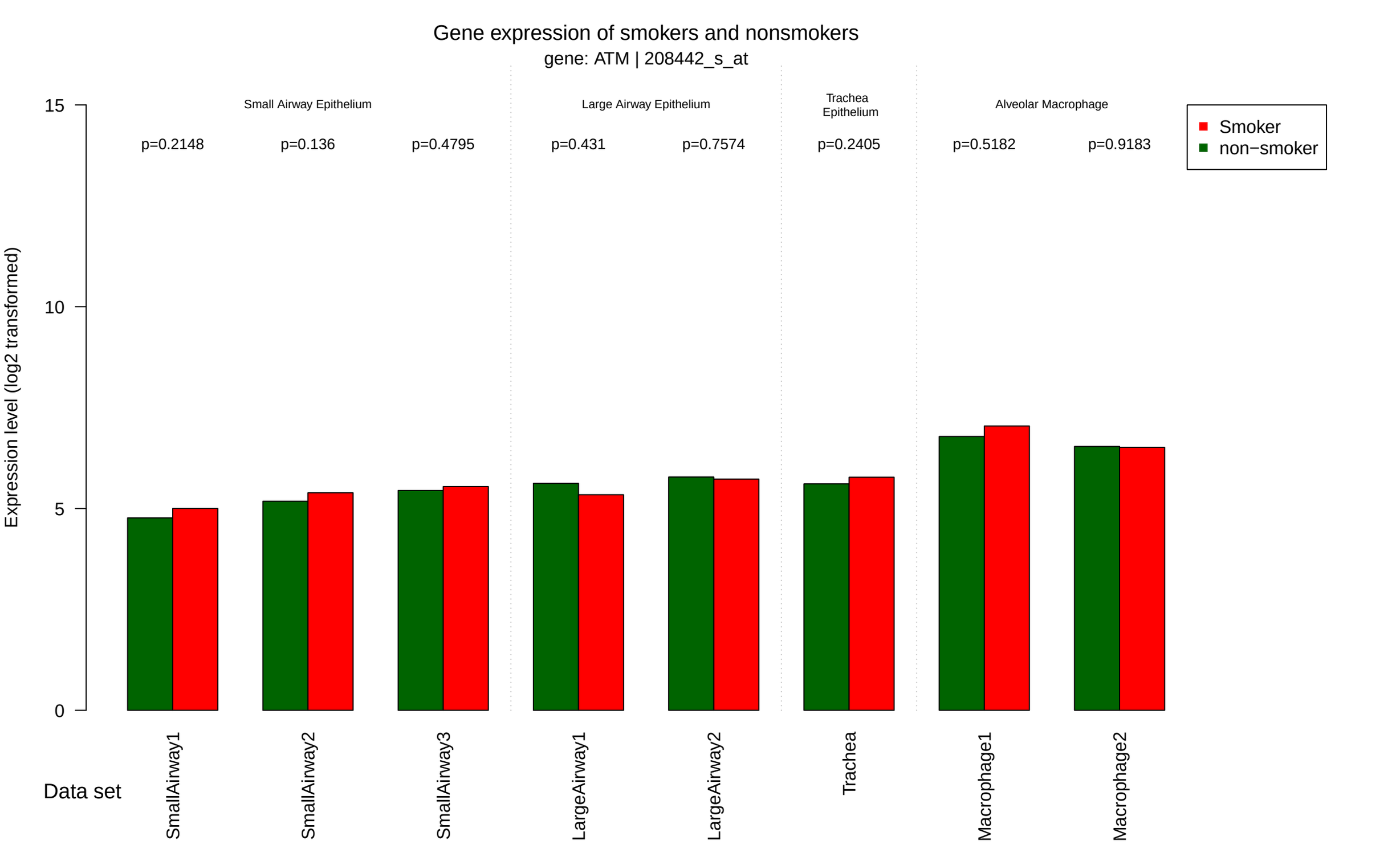
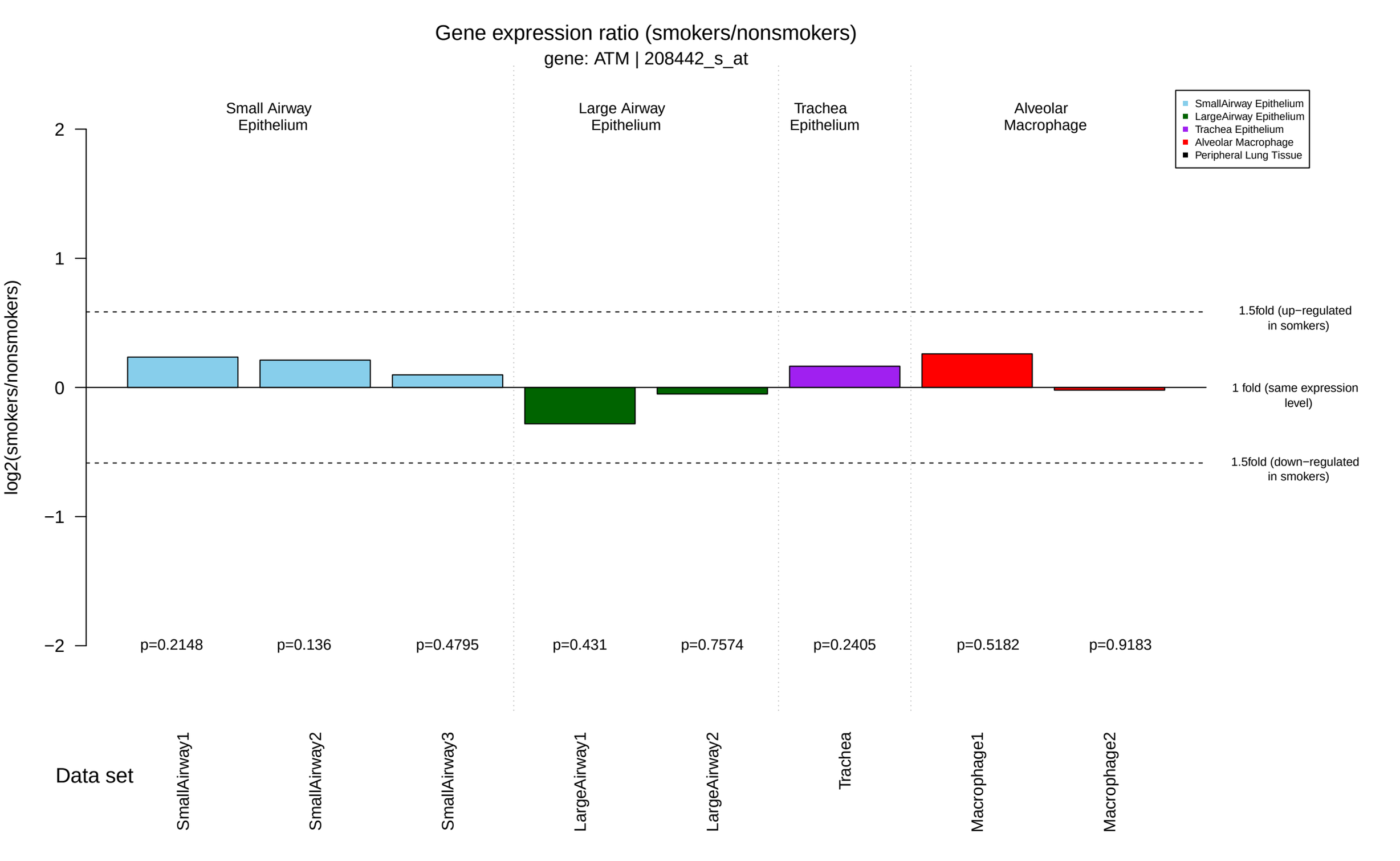
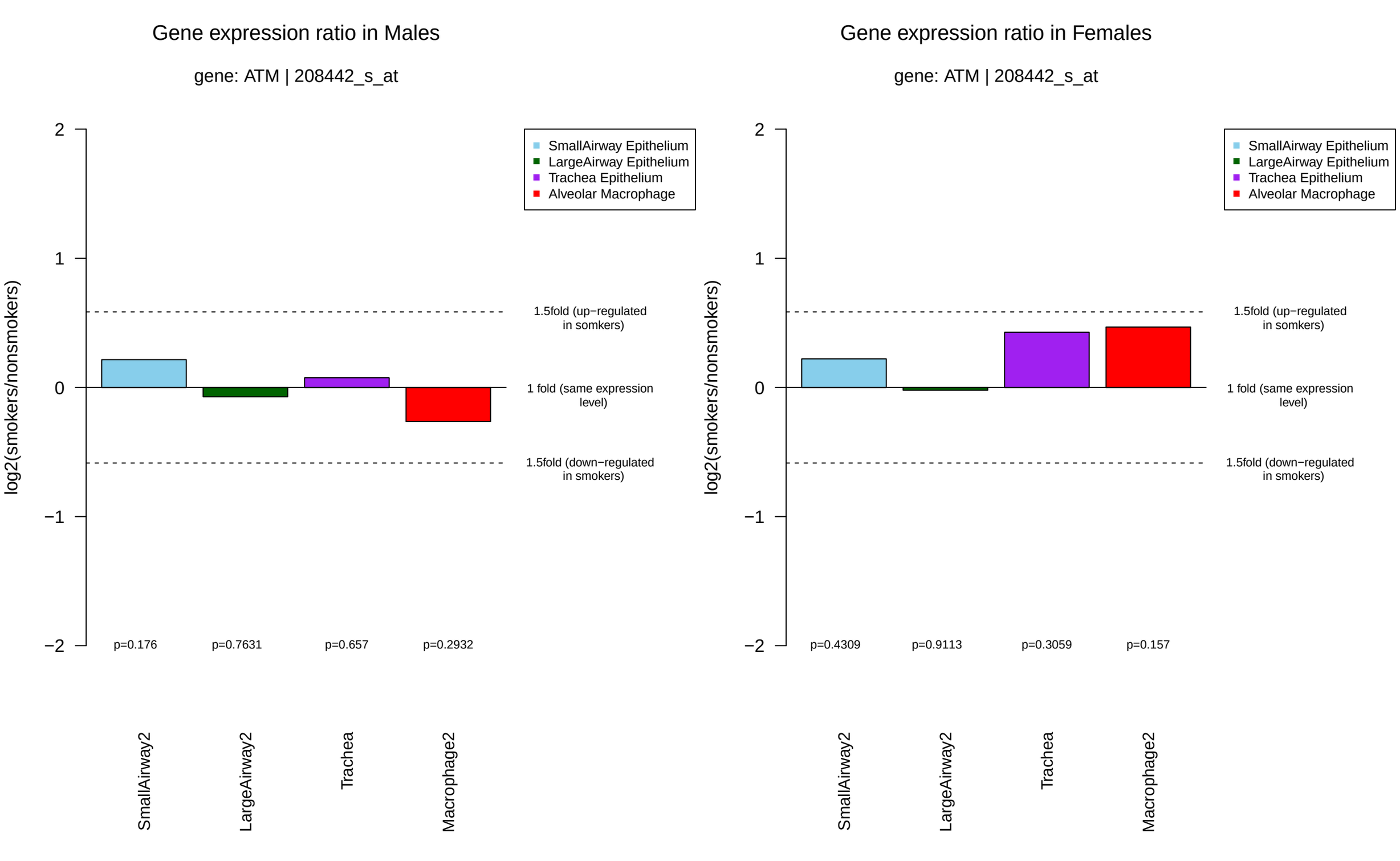
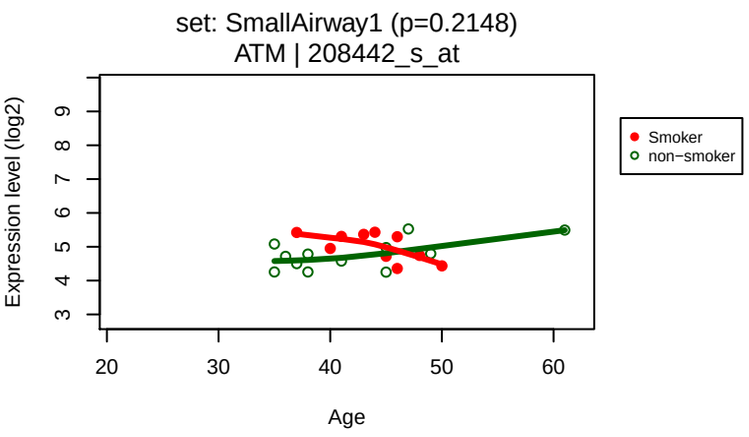
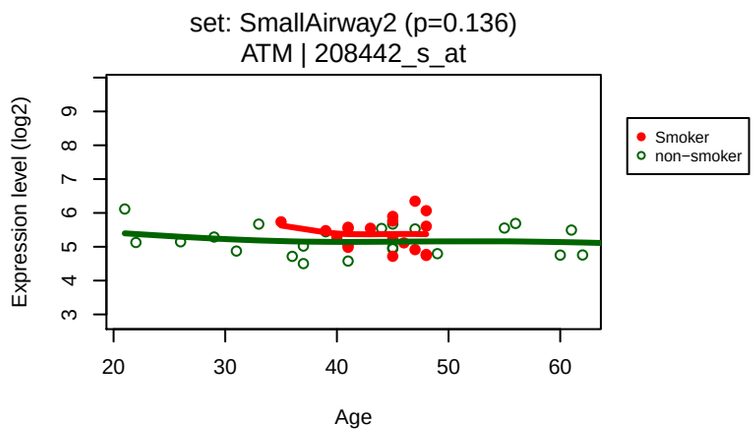
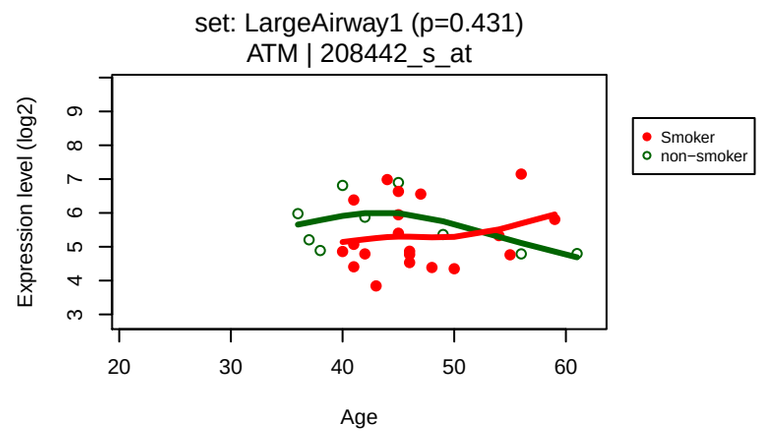

Gene expression of smokers and nonsmokers
gene: ATM | 208442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2148
p=0.136
p=0.4795
p=0.431
p=0.7574
p=0.2405
p=0.5182
p=0.9183
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

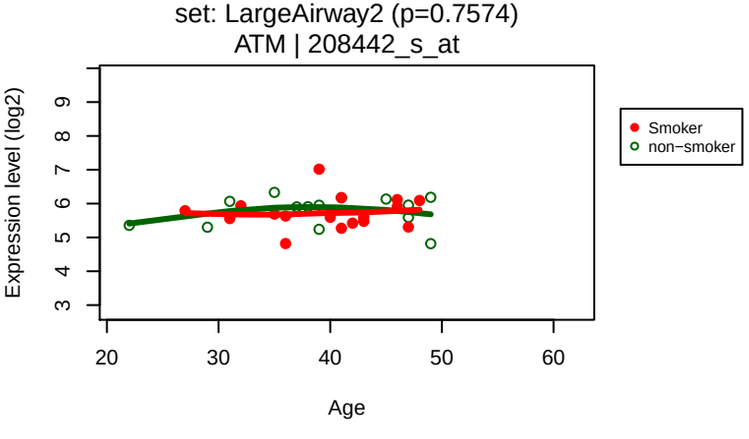

Gene expression ratio (smokers/nonsmokers)
gene: ATM | 208442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2148
p=0.136
p=0.4795
p=0.431
p=0.7574
p=0.2405
p=0.5182
p=0.9183
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

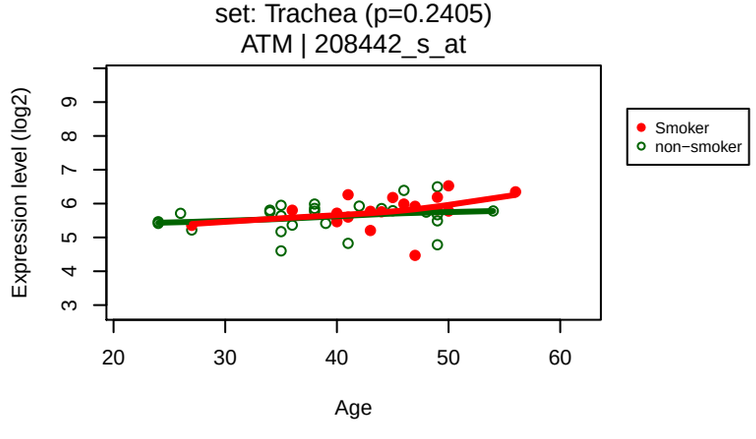

Gene expression ratio in Males
gene: ATM | 208442_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.176
p=0.7631
p=0.657
p=0.2932
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATM | 208442_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4309
p=0.9113
p=0.3059
p=0.157
SmallAirway2
LargeAirway2
Trachea
Macrophage2

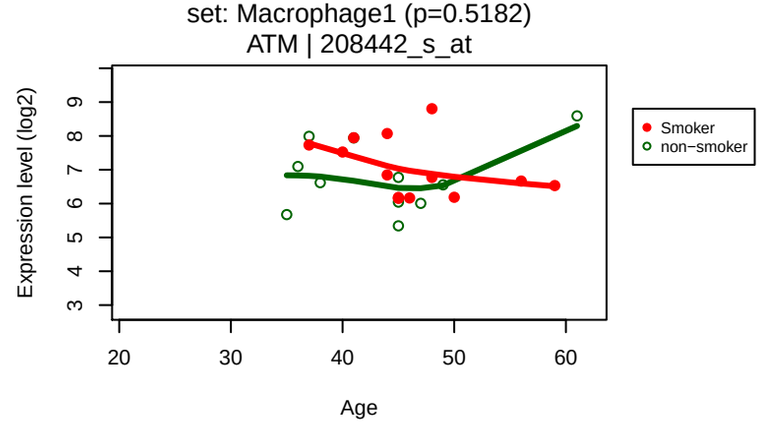

set: SmallAirway1 (p=0.2148)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker

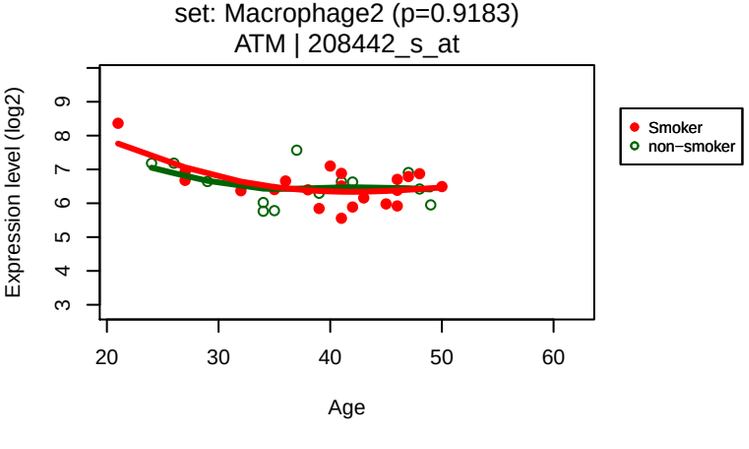

set: SmallAirway2 (p=0.136)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker

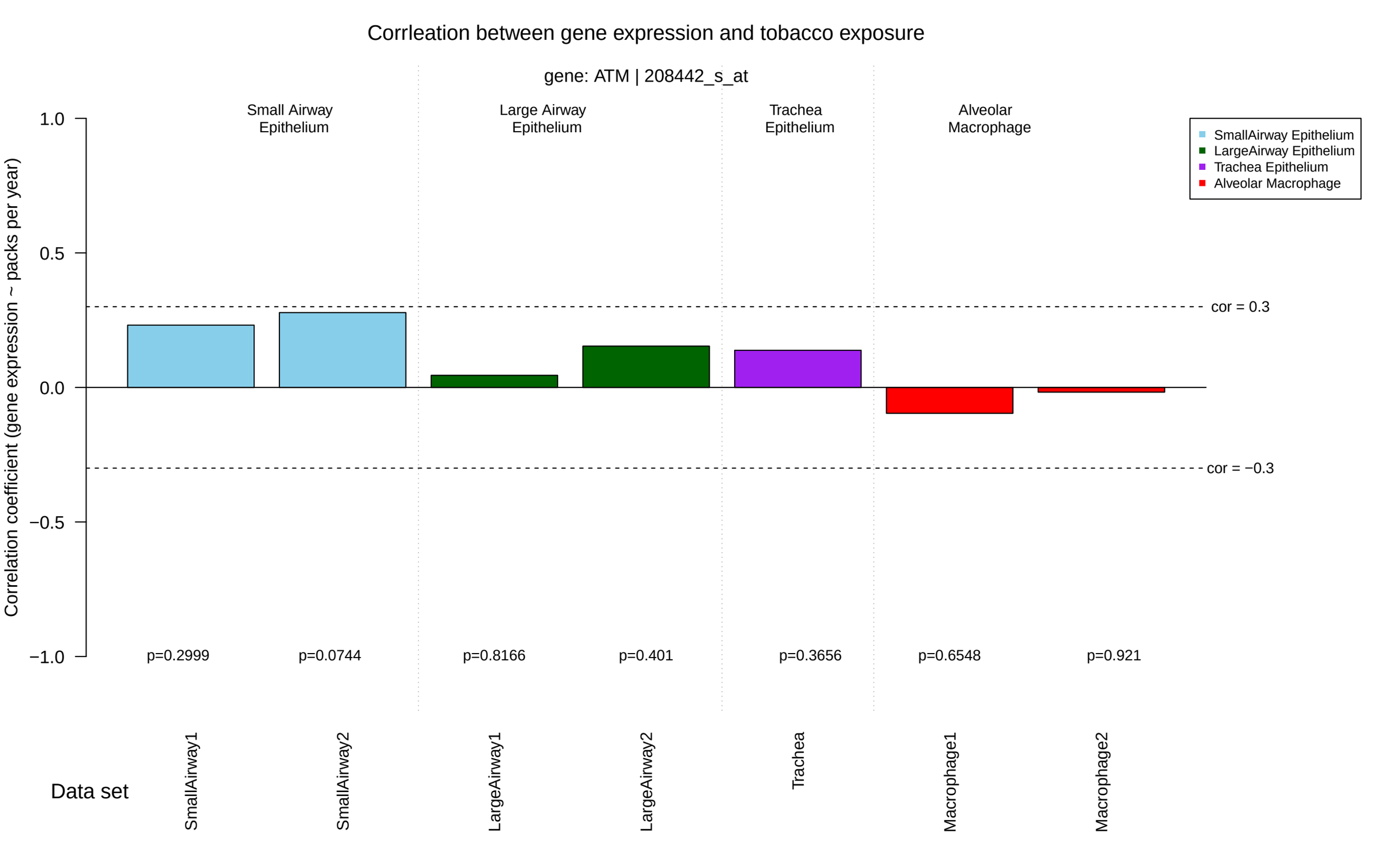

set: LargeAirway1 (p=0.431)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7574)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2405)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5182)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9183)
ATM | 208442_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATM | 208442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2999
p=0.0744
p=0.8166
p=0.401
p=0.3656
p=0.6548
p=0.921
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2